

BRIDGE LAB – FULL STACK ASSIGNMENT (14 OCT 2025)

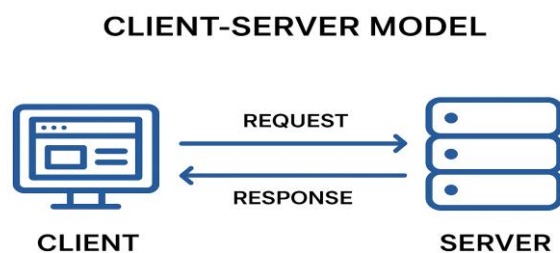
1) Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend --> UI design / User View code . Example : Visit any website and see the design (GLA PORTAL) , Simple HTML , CSS , Javascript PAGE

Backend --> Working code (means the code by which the whole website works) . Example : Working Code , Nodejs

Full -Stack --> Frontend & Backend (MERN STACK – MongoDB , Express , React , Nodejs)

2) Create a simple diagram showing how the client-server model works in web architecture.



3) Describe how a browser requests and displays a web page from a web server.

- Browser find's the server's IP address
- Browser asks for web page
- Server sends requested HTML
- Browsers processes the Html,css etc page

4) Identify and list the tools required to set up a web development environment. Explain the purpose of each.

- Text Editor --> Use to write code (VS CODE , INTELLIJI IDE etc)
- Web Browser --> Use to run the code and view web page(chrome , etc)
- Git Hub --> Use to share code with other teammates
- Database --> Use to store data (MongoDB , MySQL , etc)
- Deployment Tools --> Use to share the website globally so that other use can also use it.

5) Explain what a web server is and give examples of commonly used servers.

Web sever is a kind of software use to send web pages to the browsers over the internet . It helps in handling the requests , processes them and send back to the requested content

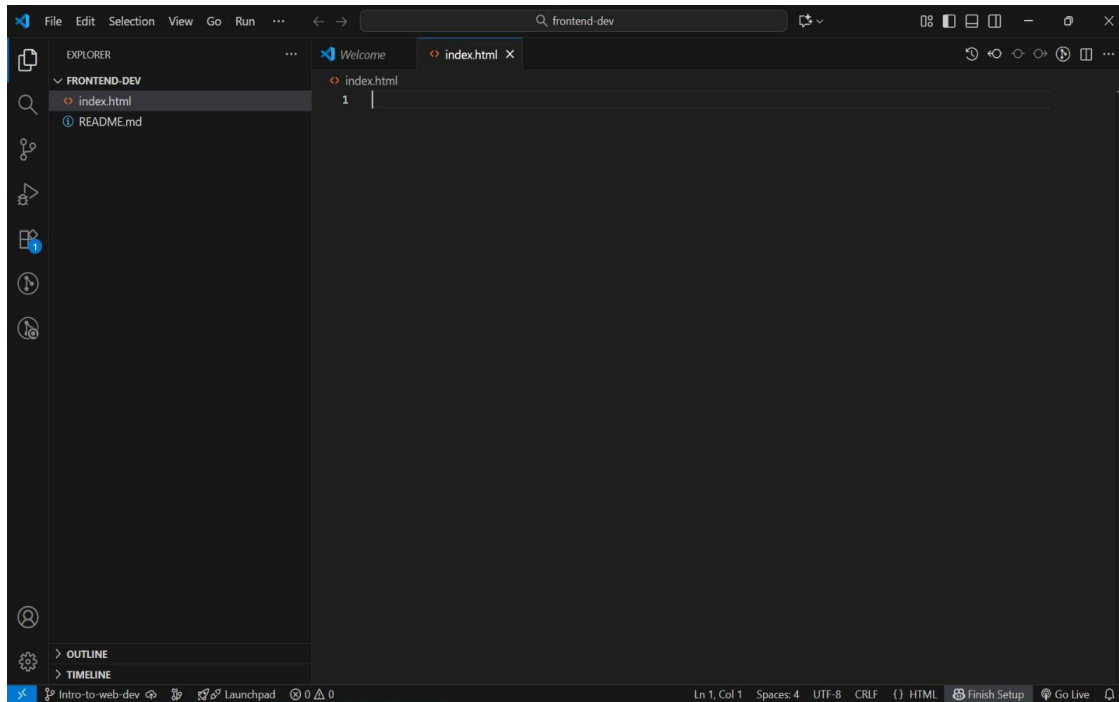
6) Define the roles of a frontend developer, backend developer, and database administrator in a project.

Frontend Developer --> Designs and implement the user Interface using HTML , CSS , JAVASCRIPT , REACT etc of the website

Backend Developer --> Build the working logic of website using NODEJS , MONGODB etc and maintain that .

Database administrator --> Manages and optimize the database for efficient data storage

7) Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.



8) Explain the difference between static and dynamic websites. Provide an example of each.

Static Websites --> Websites with fixed content means no changes will be occur after deploy unless the developer do it manually (A simple portfolio page using html,css ,js)

Dynamic Websites --> Websites which content changes as per requirement of user input or data from server (Social Media platform using MERN)

9) Research and list five web browsers. Explain how rendering engines differ between them.

- **Google Chrome:** Uses **Blink**, optimized for speed and performance.
- **Mozilla Firefox:** Uses **Gecko**, prioritizes web standards and privacy.
- **Apple Safari:** Uses **WebKit**, optimized for Apple hardware and energy efficiency.
- **Microsoft Edge:** Uses **Blink** (Chromium-based), combining Chrome's speed with Windows integration.
- **Opera:** Uses **Blink**, Chromium-based with unique features like a built-in VPN.

10) Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

